8.

①

Thm 8.3.

$$\text{WLLN:} \quad X_1, X_2 \dots \overset{iid}{\sim} (\mu, \sigma^2) \Rightarrow \bar{X}_n \xrightarrow{P} \mu$$

$\sigma^2 < \infty$

Thm 5.13

chebychev's 不等式 $\forall r > 0$, $P(g(x) \geq r) \leq \dfrac{E[g(x)]}{r}$

Thm 7.7-2

$$P(|\bar{X}_n - \mu| \geq \varepsilon) = P((\bar{X}_n - \mu)^2 \geq \varepsilon^2) \leq \frac{E[(\bar{X}_n - \mu)^2]}{\varepsilon^2} = \frac{Var[\bar{X}_n]}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2} \xrightarrow[n \to \infty]{} 0$$

$$\Leftrightarrow P(|\bar{X}_n - \mu| < \varepsilon) = 1 - P(|\bar{X}_n - \mu| \geq \varepsilon) \geq 1 - \frac{\sigma^2}{n\varepsilon^2} \xrightarrow[n \to \infty]{} 1$$

Eg. 8.4.

$$P(|S_n^2 - \sigma^2| \geq \varepsilon) \leq \frac{E[(S_n^2 - \sigma^2)^2]}{\varepsilon^2} = \frac{Var[S_n^2]}{\varepsilon^2} \xrightarrow{n \to \infty} 0 \quad \text{if } Var[S_n^n] \to 0$$

Ex8.14 Delta method.

Taylor expansion of $g(Y_n)$ around $Y_n = \theta$:

①

$$g(Y_n) = g(\theta) + g'(\theta)(Y_n - \theta) + \text{Remainder}$$

剩余項

$Y_n \to \theta \Rightarrow \text{Remainder} \to 0$

$$g'(\theta)(Y_n - \theta) = g(Y_n) - g(\theta)$$

$$\sqrt{n}\, g'(\theta)(Y_n - \theta) = \sqrt{n}\left[g(Y_n) - g(\theta)\right]$$

$$\because \sqrt{n}(Y_n - \theta) \xrightarrow{D} N(0, \sigma^2)$$

$$\Rightarrow g'(\theta) \cdot \sqrt{n}(Y_n - \theta) \xrightarrow{D} N(0, [g'(\theta)]^2 \sigma^2)$$

②

$$g(Y_n) = g(\theta) + g'(\theta)(Y_n - \theta) + \frac{g''(\theta)}{2!}(Y_n - \theta)^2 + \text{Remainder}$$

$$= g(\theta) + 0 + \frac{g''(\theta)}{2!}(Y_n - \theta)^2 + \text{Remainder}$$

$$g(Y_n) - g(\theta) = \frac{g''(\theta)}{2!}(Y_n - \theta)^2$$

$$\because \left[\frac{\sqrt{n}(Y_n - \theta)}{\sigma}\right]^2 \xrightarrow{D} \chi_1^2$$

$$\frac{n}{\sigma^2}\left(g(Y_n) - g(\theta)\right) = \frac{n}{\sigma^2} \cdot \frac{g''(\theta)}{2!}(Y_n - \theta^2) + \text{Remainder}$$

$$n\left[g(Y_n) - g(\theta)\right] = \sigma^2 \cdot \frac{g''(\theta)}{2!}\left(\frac{\sqrt{n}}{\sigma}(Y_n - \theta)\right)^2 \xrightarrow{D} \sigma^2 \frac{g''(\theta)}{2!}\chi_1^2$$

8.11. CLT.

③

$\forall |t|<h,\quad M_{\frac{\sqrt{n}(\bar{X}-\mu)}{\sigma}}(t) \longrightarrow e^{t^2/2} \quad (n\to\infty) \quad \Rightarrow \quad \frac{\sqrt{n}(\bar{X}-\mu)}{\sigma} \sim N(0,1)$

$Y_i = \frac{X_i-\mu}{\sigma}$ $\qquad$ Y_i の MGF: $M_Y(t)$ 存在する. $\forall |t|<\sigma h$

$$\frac{1}{\sqrt{n}}\sum_{i=1}^{n} Y_i = \frac{1}{\sqrt{n}}\sum_{i=1}^{n}\frac{X_i-\mu}{\sigma} = \frac{1}{\sqrt{n}}\cdot\frac{n\bar{X}_n-n\mu}{\sigma} = \frac{\sqrt{n}(\bar{X}_n-\mu)}{\sigma}$$

$$M_{\frac{\sqrt{n}(\bar{X}-\mu)}{\sigma}}(t) = M_{\frac{1}{\sqrt{n}}\sum_{i=1}^{n} Y_i}(t) \underset{\text{Thm 5.10}}{=} M_{\sum_{i=1}^{n} Y_i}\left(\frac{t}{\sqrt{n}}\right) \underset{\text{Thm 5.8}}{=} \left(\underline{M_Y\left(\frac{t}{\sqrt{n}}\right)}\right)^n \quad ⊛$$

↳ Taylor Expansion する △

△ $M_Y\left(\frac{t}{\sqrt{n}}\right) = E\left[e^{\frac{t}{\sqrt{n}}Y}\right]$ $\qquad$ ← supp 1.6 e^x の Taylor expansion

$$= E\left[\sum_{k=0}^{\infty}\frac{\left(\frac{t}{\sqrt{n}}Y\right)^k}{k!}\right] = \sum_{k=0}^{\infty} EY^k\cdot\frac{\left(\frac{t}{\sqrt{n}}\right)^k}{k!} = \sum_{k=0}^{\infty} M_Y^{(k)}(0)\frac{(t/\sqrt{n})^k}{k!}$$

← Thm 1.4. $\qquad$ → $\forall\left|\frac{t}{\sqrt{n}}\right|<\sigma h \Rightarrow t<\sqrt{n}\sigma h$, $M_Y^{(k)}(0)$ 存在する.

$M_Y^{(0)}=1 \to EY^0=1$

$M_Y^{(1)}=0 \to EY=0$

$M_Y^{(2)}=1 \to EY^2 = Var(Y)-(EY)^2 = 1$

$$= M_Y^{(0)}(0)\frac{(t/\sqrt{n})^0}{0!} + M_Y^{(1)}(0)\cdot\frac{(t/\sqrt{n})}{1!} + M_Y^{(2)}(0)\frac{(t/\sqrt{n})^2}{2!} + \cdots$$

$$= 1 + 0 + \frac{(t/\sqrt{n})^2}{2!} + R_Y\left(\frac{t}{\sqrt{n}}\right)$$

↳ $\sum_{k=3}^{\infty} M_Y^{(k)}(0)\frac{(t/\sqrt{n})^k}{k!}$

$$\lim_{n\to\infty}\frac{R_Y(t/\sqrt{n})}{(t/\sqrt{n})^2} = 0$$

④.

$$R_Y\left(\frac{t}{\sqrt{n}}\right) = \sum_{k=3}^{\infty} M_Y^{(k)}(0)\,\frac{(t/\sqrt{n})^k}{k!}$$

$$\lim_{n\to\infty} \frac{R_Y(t/\sqrt{n})}{(t/\sqrt{n})^2} = 0 \quad \Rightarrow \quad \lim_{n\to\infty} \frac{R_Y(t/\sqrt{n})}{(1/\sqrt{n})^2} = \lim_{n\to\infty} n\,R_Y(t/\sqrt{n}) = 0$$

$t=0$: $R_Y(0/\sqrt{n}) = 0$

$t\neq 0$: $\lim_{n\to\infty} n\cdot R_Y(t/\sqrt{n}) = 0$

$$\circledast \quad \lim_{n\to\infty} \left\{M_Y\left(\frac{t}{\sqrt{n}}\right)\right\}^n = \lim_{n\to\infty}\left[1 + \frac{(t/\sqrt{n})^2}{2!} + R_Y\left(\frac{t}{\sqrt{n}}\right)\right]^n$$

$$= \lim_{n\to\infty}\left[1 + \frac{t^2}{n\cdot 2!} + R_Y\left(\frac{t}{\sqrt{n}}\right)\right]^n$$

$$= \lim_{n\to\infty}\left[1 + \frac{1}{n}\left(\frac{t^2}{2!} + n\cdot R_Y\left(\frac{t}{\sqrt{n}}\right)\right)\right]^n$$

$$= e^{\frac{t^2}{2}} \quad \#$$

$\leftarrow$ $N(0,1)$ の MGF.

$$\lim_{n\to\infty} a_n = a \quad \Rightarrow \quad \lim_{n\to\infty}\left(1 + \frac{a_n}{n}\right)^n = e^a$$

$\#$

$$\lim_{n\to\infty} \underbrace{\frac{t^2}{2} + nR_Y\left(\frac{t}{\sqrt{n}}\right)}_{a_n} = \underbrace{\frac{t^2}{2}}_{a} \quad \Rightarrow \quad \lim_{n\to\infty}\left[1 + \frac{1}{n}(a_n)\right]^n = e^a$$